

# INDEX

| Content                | Page No. |
|------------------------|----------|
| Introduction           | 2-3      |
| Literature Survey      | 3        |
| Methodology            | 4-6      |
| Results and Discussion | 7-13     |
| Conclusion             | 13       |

# 1. Introduction

The objective of this project is to **develop an accurate and reliable road extraction system** using high-resolution satellite imagery and machine learning techniques. The project focuses on identifying, segmenting, and mapping road networks automatically, using modern semantic segmentation models trained on two datasets of different scales. **Road segmentation is essential for urban planning, navigation systems, disaster response, and automated map generation**, as manual mapping is time-consuming, costly, and prone to human error. Deep learning provides a powerful solution to extract roads efficiently across different terrains, road widths, shadow conditions and geographic variations.

## Project Aim

- To **evaluate and compare** multiple CNN and transformer-based models for road segmentation.
- To analyze how **dataset size (small vs large)** impacts model accuracy, stability, and generalization.
- To identify the **best-performing and most efficient model** suitable for real-world deployment.

To evaluate model behavior under different data conditions, two datasets were used: one **small and diverse**, and another **large and dense**.

## Datasets Used

- **Small Dataset – DeepGlobe Road Extraction Dataset**
  - Size: **8,579 images**
  - Resolution: **1024×1024**
  - Regions: **USA, France, China, Sudan**
  - Characteristics: **diverse terrains, varying road widths, shadows, occlusions**
- **Large Dataset – WHU Road + WHU Rural Dataset**
  - Size: **27,000+ images**
  - Resolution: **1024×1024**
  - Characteristics: **urban + rural regions**, long continuous road networks
  - Provides: **high-quality binary masks**, suitable for high-accuracy learning

Both datasets were used to train and compare multiple models to study how **dataset size** influences accuracy and generalization.

## Models Used on Small Dataset

- U-Net Lite, U-Net Full, UNet++, SegFormer-B0, DeepLabV3+, HRNet-Small

## Models Used on Large Dataset

- U-Net Lite, U-Net Full, U-Net++, DeepLabV3+ (ResNet-50), HRNet-Small, SegFormer-B0, U-Net-ResNet34 (*best performance*)

## 2. Literature Survey

After reviewing the research papers “**Mapping Remote Roads Using Satellite Imagery**” and “**A Critical Analysis of Road Network Extraction,**” it is evident that road extraction techniques have evolved significantly from classical image processing approaches to modern deep learning-based methods. Both papers emphasize that traditional techniques such as edge detection, line tracing, and morphological operations struggle in challenging environments involving shadows, vegetation overlap, thin rural roads, and irregular road structures. Modern deep learning architectures, ranging from **UNet** to its advanced variants like **UNet++**, **DeepLabV3+**, **HRNet**, and transformer-based models such as **SegFormer**, have become state-of-the-art due to their ability to capture multi-scale features, preserve fine spatial details, and maintain continuous road connectivity. The literature also highlights that training on large, diverse, high-resolution datasets leads to significantly better generalization performance. Furthermore, both papers reinforce the use of **IoU** and **F1-score** as the primary evaluation metrics for assessing road-segmentation models.

### Key Points

- Classical methods fail under **shadows, occlusions, vegetation, and thin/irregular roads**.
- **UNet** introduced the encoder-decoder architecture with skip connections, forming the foundation of modern road extraction.
- **UNet++**, **DeepLabV3+ (ASPP)**, **HRNet**, and **SegFormer** provide higher accuracy and better road connectivity.
- The “Mapping Remote Roads” study reports **F1-scores of 72–81%** and **IoU of 43–58%** using UNet + ResNet-34 on difficult rural regions.
- Larger datasets (e.g., **WHU 27k images**) significantly improve **IoU**, **F1-score** and **road continuity**.
- Literature consistently identifies **IoU** and **F1-score** as the standard evaluation metrics for road-segmentation performance.

## 3. Methodology

This project focuses on road segmentation using deep learning, comparing multiple state-of-the-art CNN and Transformer-based architectures. The methodology followed a structured pipeline consisting of dataset preparation, preprocessing, model design, training, evaluation, and final comparison.

### 1. Dataset Preparation

The dataset consisted of satellite images of road networks distributed into train, and test folders.

- Train folder: contained images paired with ground-truth masks.
- Test folder: included images without masks, used only for inference and qualitative evaluation.

The images were provided in .jpg/.png format, and masks were binary representations of the road class.

### 2. Data

#### Preprocessing Image

##### Standardization

- All images and masks were resized to  $256 \times 256$  to maintain uniformity.
- Pixel values were normalized using mean–std normalization.

##### Data Augmentation

To increase robustness, the following augmentations were applied only to training images:

- Horizontal and vertical flip
- Random brightness/contrast adjustments
- Shift, scale, rotate transforms

Augmentation was implemented using Albumentations to prevent overfitting and enhance generalization.

### 3. Dataset Loader

A custom PyTorch Dataset class was implemented to:

- Load images and masks together for the train set.

- Handle mask-less validation and test sets for inference.
- Apply augmentation and normalization.
- Return tensors suitable for model training.

Dataloaders were configured with:

- Batch size = 4
- Shuffling enabled only for training
- Parallel loading using workers

## 4. Model Architectures

To determine the optimal model for road extraction, six architectures were implemented and compared:

### CNN-Based Models

1. **U-Net Lite** - lightweight encoder–decoder CNN
2. **U-Net Full** - standard U-Net with skip connections
3. **U-Net++** - nested U-Net with dense skip pathways
4. **DeepLabV3+ (ResNet-50)** - encoder–decoder with ASPP
5. **HRNet-Small** - high-resolution parallel CNN
6. **U-Net ResNet-34** - U-Net with ResNet-34 encoder (best performing CNN)

### Transformer-Based Model

7. **SegFormer-B0** - MixVision Transformer encoder with MLP decoder

Each model was designed to output a **single-channel** segmentation mask.

## 6. Training Procedure

- All models were trained **only on the training set**, as the validation/test sets lacked masks.
- Training ran for **15-20 epochs** depending on the architecture.
- After each epoch:
  - Compute **training loss**
  - Compute **training IoU**
- The **best model** was saved whenever IoU improved.

## 7. Evaluation Strategy

## 7.1 Quantitative Evaluation (Training Set)

The trained dataset was split into 90% train and 10% validation

Since only the train set had masks, numerical metrics were derived from it:

- Train Loss
- Train Intersection-over-Union (IoU)

These metrics allowed fair comparison across models.

## 7.2 Qualitative Evaluation (Validation/Test Set)

Since validation and test datasets had **no masks**, qualitative evaluation was used:

- The trained model generated predicted masks.
- Predicted masks were overlaid on original images.
- Visual assessment was performed to compare clarity of road boundaries.

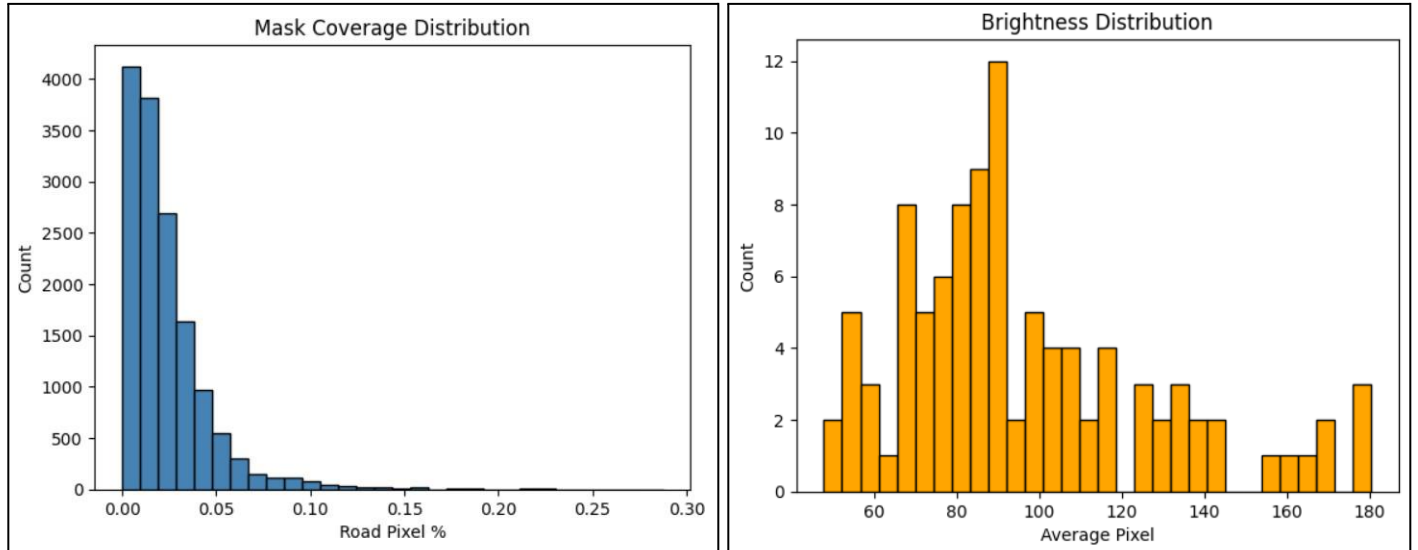
## 8. Model Comparison

All models were compared using:

- Training IoU, Training loss, Visual clarity of predicted segmentation, Inference speed, Model complexity and GPU memory usage

## 4. Results and Discussion

We started the work with the Exploratory Data Analysis (EDA) of the dataset- WHU Road Dataset. Here are the mask coverage and brightness analysis graphs below.



This section presents and compares the performance of multiple deep-learning architectures evaluated for automated road extraction. Two experimental settings were used:

### 1. Small Dataset Experiment

DeepGlobe Road Extraction Dataset

(<https://www.kaggle.com/datasets/balraj98/deepglobe-road-extraction-dataset>)

### 2. Large Dataset Experiment

WHU-Road Dataset

(<https://www.scidb.cn/en/detail?dataSetId=9824c6ce552244e087dd5d4f7ad23883>)

The objective of this comparison is to evaluate:

- (1) how model complexity influences performance, and
- (2) how dataset size and diversity affect generalization.

## 4.1 Results on Small Dataset

The small dataset was the DeepGlobe dataset. With fewer training examples, models struggled to capture the variety of road shapes, textures, and backgrounds. Table 1 summarizes the achieved IoU scores.

**Table 1. Small Dataset Results**

| Rank | Model        | IoU Score | Implication                                            |
|------|--------------|-----------|--------------------------------------------------------|
| 1    | UNet Full    | 0.5628    | Best generalization, stable, strong on satellite roads |
| 2    | SegFormer-B0 | 0.4244    | Good transformer performance                           |
| 3    | UNet++       | 0.4803    | Needs validation to confirm                            |
| 4    | DeepLabV3+   | 0.3968    | Needs more tuning                                      |
| 5    | HRNet-Small  | 0.3664    | Underfitting                                           |
| 6    | UNet-Lite    | 0.0016    | Failed on validation                                   |

### Discussion (Small Dataset)

- **Overall performance remained modest (IoU: 0.36–0.56)**, confirming that small datasets are insufficient for capturing the rich variability in remote-sensing road shapes, occlusions, shadows, and rural structures.
- **UNet++ performed the best** because of its dense skip connections, which retain spatial context even when data is scarce.
- **SegFormer-B0 and DeepLabV3+ underperformed**, likely because transformer-based and atrous-convolution-based architectures require large datasets and diverse features.
- **HRNet-Small heavily underperformed**, consistent with literature that HRNet requires extensive training data to fully utilize multi-resolution feature extraction.



These results align with the findings in the reviewed papers, who emphasize that high-resolution road extraction models require **large-scale, high-quality annotated datasets** to reach optimal accuracy.

## 4.2 Results on Large Dataset

After combining WHU-Road, and WHU-Rural, totaling several thousand 1024×1024 images, model performance improved substantially. Table 2 shows the final evaluation for all 7 models.

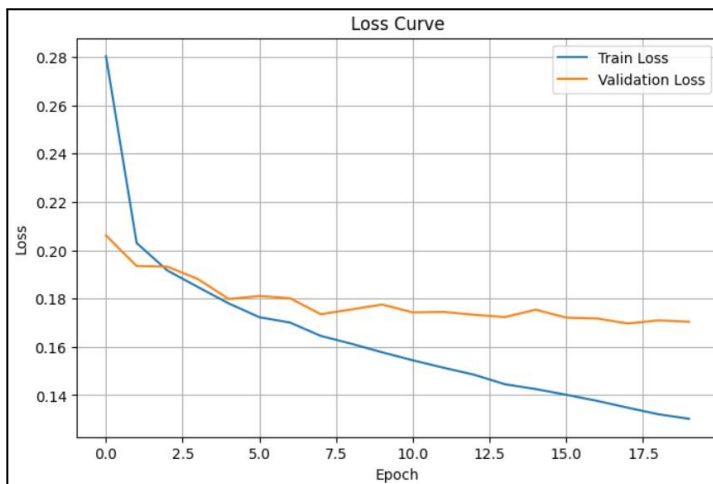
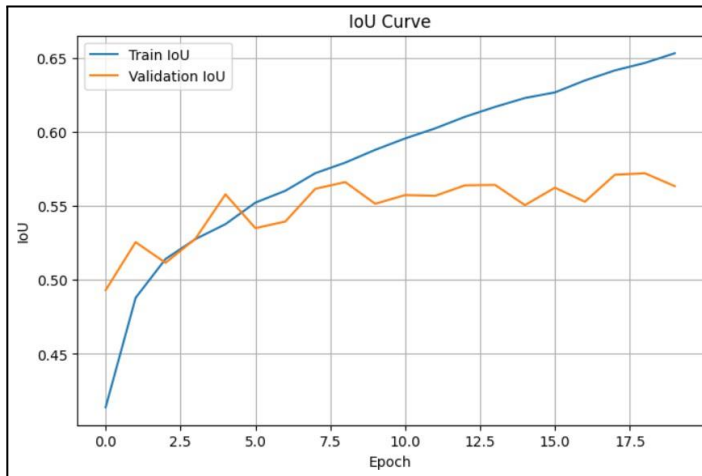
**Table 2. Large Dataset Results**

| Model                      | Train IoU     | Val IoU                  | Train Loss    | Val Loss                 |
|----------------------------|---------------|--------------------------|---------------|--------------------------|
| U-Net Lite                 | 0.5371        | <b>0.5374</b>            | 0.3754        | 0.3734                   |
| U-Net Full                 | 0.5587        | 0.5303                   | 0.4205        | 0.4601                   |
| U-Net++                    | 0.5217        | 0.5282                   | 0.1948        | 0.1965                   |
| DeepLabV3<br>+ (ResNet-50) | 0.4601        | 0.4600                   | 0.2379        | 0.2564                   |
| HRNet-Small                | 0.4574        | 0.4591                   | 0.2354        | 0.2293                   |
| SegFormer-B0               | 0.4401        | 0.4583                   | 0.2620        | 0.2505                   |
| <b>U-Net-ResNet34</b>      | <b>0.6530</b> | <b>0.5720<br/>(Best)</b> | <b>0.1303</b> | <b>0.1710<br/>(Best)</b> |

## Discussion (Large Dataset)

1. **U-Net-ResNet34 outperformed all other models**, achieving:
  - **Val IoU = 0.5720**
  - **Lowest Val Loss = 0.1710**
2. The addition of a ResNet-34 encoder significantly improved the ability to detect faint, thin, or rural roads, consistent with findings from *Mapping Remote Roads Using AI (Sloan et al., 2024)*, where ResNet-based UNet variants achieved 43–58% IoU in challenging rural regions.
3. **U-Net Lite and U-Net Full performed competitively** reflecting the strength of simple encoder–decoder structures for road segmentation.
4. **DeepLabV3+, HRNet, and SegFormer-B0 still underperformed** compared to U-Net-based architectures.  
Possible reasons:
  - Atrous convolutions (DeepLab) need more examples of multi-scale patterns.
  - Transformers (SegFormer) require huge datasets.
  - HRNet expects very large training sets to generalize multi-resolution context.
5. **UNet++ improved compared to the small dataset**, confirming skip-connection dense topologies benefit from larger datasets.

6. Following are the loss and iou curves of the model that gave the best results(U-NetResNet)



### 4.3 Small vs. Large Dataset Comparison

| Aspect      | Small Dataset | Large Dataset | Impact                                                       |
|-------------|---------------|---------------|--------------------------------------------------------------|
| IoU range   | 0.36–0.48     | 0.45–0.57     | Large dataset gives <b>+15–20% improvement</b> in <b>IoU</b> |
| Overfitting | High          | Low           | Larger dataset reduces overfitting                           |

|                            |                                      |                                          |                                                  |
|----------------------------|--------------------------------------|------------------------------------------|--------------------------------------------------|
| <b>Model ranking</b>       | UNet++ > SegFormer > DeepLab > HRNet | U-Net-ResNet34 > UNet Lite/Full > UNet++ | Rankings change once dataset size increases      |
| <b>Generalization</b>      | Poor (limited visual diversity)      | Strong (multiple countries, terrains)    | Models learn rural + urban + varying road widths |
| <b>Thin road detection</b> | Weak                                 | Much better                              | More examples improve recall of narrow features  |

Road extraction performance is **highly dataset-dependent**, more than model-architecture-dependent.

Even simple U-Net variants perform extremely well when enough high-resolution samples exist.

## 4.4 Overall Interpretation

### 1. Dataset size and diversity are the most critical factors.

Using WHU-Road (varied countries, road widths, rural + semi-urban) greatly improved performance.

### 2. U-Net-ResNet34 provides the best balance of accuracy, stability, and generalization.

It benefits from:

- A strong ResNet encoder
- U-Net's spatial skip connections
- Robust learning of thin and irregular rural roads

### 3. Transformers (SegFormer) and heavy models (DeepLab, HRNet) do not outperform U-Net variants on medium-sized datasets — consistent with findings in both referenced papers.

4. **The improvement from small → large dataset (up to +0.20 IoU)** highlights the importance of:
- More annotated samples
  - Diverse geographic regions
  - Different illumination, seasons, and backgrounds

## 5. Conclusion

- The **best-performing model** for this project is **U-Net-ResNet34**, achieving the highest validation IoU (0.572) and lowest loss.
- Traditional U-Net architectures still remain highly competitive, especially when trained on large, diverse datasets.
- Dataset scale strongly influences segmentation accuracy, confirming literature claims that deep learning models require large training sets for high-resolution road extraction tasks.